

INTERVIEW PREPARATION

1) Constructor :

- * constructor is special type of function inside class.
- * constructor is used to initialise the data member of new object.
- * constructor name should be same as their class.
- * constructor doesn't have any parameter.
- * Also constructor doesn't have any return type.
- * Constructor call itself at the time of object creation.

2) Default Constructor :

- * in C# constructor is special member function.
- * when we don't define any constructor in class that time by default call default constructor.
- * default constructor doesn't have any parameters.

3) Parameterized Constructor :

- * A constructor which has parameters is called parameterized constructor.
- * It is used to provide different values to distinct objects.

4) Copy Constructor

- * A constructor that initialize as a new object by copying data from another object.

5) Static Constructor

- * when we create static constructor we need static keyword.
- * static constructor can only initialize static variable.
- * in static constructor we can not pass parameter.

6) Static function :

- * static function calling hum class ke name ke sath karate hai.
- * static function we define with static keyword.
- * static function only contain static variable, static member.

7) Function :

- * function is block of code and reusable of code that performs a specific task.
- * we can function call of object creation.

8) function overloading :

- * Multiple method of same name in single class.
- * No need of inheritance, as it is in single class .
- * in function overloading All methods have different parameter (signature).
- * function overloading is compile time polymorphism.
- * No special keyword used in function overloading.

9) function overriding :

- * Multiple method of same name in different class.
- * Inheritance is used, as it is in different class.
- * in the function overriding in which All methods have same parameter.
- * function overriding is run time polymorphism.
- * Virtual & override keywords used in function overriding.

10) Abstract class :

- * Abstract class contains both Declaration & Definition of methods.
- * when we create Abstract class abstract keyword is used.
- * Abstract class does not support multiple inheritance.
- * Abstract class can have constructors.

11) Interface Class :

- * mostly Interface Class contain Declaration of methods.
- * when we create Interface Class Interface keyword is used.
- * Interface supports multiple inheritance.
- * Interface do not have constructors.

12) What is ref and out in c#?

- * in c#, both ref and out are used for passing arguments by reference, but they have their different behaviors.
- * the ref parameters must be initialized before pass to the method.
- * the out parameters dos not need to initialized before pass to the method.

13) What is the purpose of "using" keyword in c#?

* "using" statement is use for DISPOSE() method of the class object is called even if an exception occurs.

14) What is the "this" keyword in c#?

* in C# "this" keyword is used to refer to the CURRENT INSTANCE of the class.

15) Explain the four steps involved in the C# code compilation .

- * Source code compilation in managed code.
- * Newly created code is clubbed with assembly code.
- * The Common Language Runtime (CLR) is loaded.
- * Assembly execution is done through CLR.

******* four pillars of OOP's *******

16) Encapsulation :

- * Encapsulation is the main feature of object oriented programing.
- * Encapsulation is mechanism in which data member and method is wrapped in a single unit inside a class.
- * C# Encapsulation is can't be accessed by another class.
- * Ex. medicine capsule which covered outside.

17) Inheritance :

- * inheritance is the main feature of object oriented programing.
- * inheritance allows us to create a new class from an existing class.
- * we perform inheritance in derived class (child class).
- * there are four types of inheritance in c sharp.
- * 1. single inheritance in which there is one base class and one derived class.
- * 2. Hierarchical inheritance in which one class feature that is needed in multiple classes.
- * 3. Multilevel inheritance meanse one class is derived from another.
- * 4. C# does not support multiple inheritances of classes.

* Ex. jaise hmare papa dada pardada aur purvajo ke containt hmare undar aahe aate hai like hair style hmari height etc.

18) Abstraction :

- * Abstraction is the main feature of object oriented programing.
- * Abstraction means showing only required things and hide the BACKGROUND information.
- * in c# Abstraction is make your program secure and more structured.
- * Ex. remote of your TV. you can change the channels, volume up, volume down means you can use only functionalities and background things you don't know.

19) Polymorphism :

- * Polymorphism is the main feature of object oriented programing.
- * Polymorphism is divided into two Greek words poly means many and morph means form.
- * So Polymorphism mince ek class aur multiple methods.
- * C# Polymorphism means one name with multiple functionalities.
- * Ex. cook - jaise ek cook multiple subjiyaa bana leta hai.

******* base class - parent class *******

******* derived class - child class *******

20) Virtual and override :

- * to declare a virtual method "Virtual" keyword is used.
- * when we re-define any method in derived class that time we use virtual keyword.
- * virtual allows to user re-define method in derived class.
- * when we override a method we need to use "Override" keyword.
- * In C#, a virtual method is a method that can be overridden in a derived class.

21) Abstract Class :

- * abstract keyword is used to create abstract class.
- * abstract class contain both declaration and definition of method.
- * abstract class does not support multiple inheritance.
- * abstract class can have constructor.

- * An abstract class can contain abstract methods and non-abstract methods.

22) Interface Class :

- * interface keyword is used to create an interface class.
- * interface class contain declaration of methods.
- * interface support multiple inheritance.
- * interface do not have constructor.
- * Interfaces allow you to define a common set of functionality that multiple classes can share.
- * Interface is an abstract class that has only public abstract method.

23) Static class :

- * Static class can be created using the static keyword. It contains static data members, static constructors, and static methods. However, you cannot create objects in it.

24) Partial class :

- * Partial class can divide the functionality of one class into multiple ones. The keyword partial is used to create it.

25) Abstract class :

- * This class type is restricted and cannot be used to create objects. However, it can be accessed by inheriting it from another class.

26) Sealed class :

- * Sealed class can be instantiated but not inherited by any class. The keyword sealed is used to restrict inheritance.
- * Sealed always with override keyword.

27) Properties :

- * using properties we can access private variable outside of the class. using get and set properties.
- * properties doesn't have storage location.
- * properties can be read-only or write-only.
- * properties are extension of fields and accessed like fields.

28) Dynamic Type :

- * dynamic that avoids compile-time type checking.
- * in dynamic we don't need to type casting.

- * A dynamic type variables are defined using the dynamic keyword.
- * Dynamic types change types at run-time based on the assigned value.
- * The dynamic type variables is converted to other types implicitly.

29) Array And Array Types :

- * array is a group of similar types of elements in one variable.
- * In C#, array index starts from 0. We can store only fixed set of elements in C# array.
- * there are three types of array in c# Single Dimensional, Multidimensional and jagged array.
- * To create single dimensional array, you need to use square brackets [] after the type.
- * The multidimensional array is also known as rectangular arrays in C#.
- * It can be two dimensional or three dimensional.
- * To create multidimensional array, we need to use comma inside the square brackets.
- * In C#, jagged array is also known as "array of arrays".
- * The element size of jagged array can be different.
- * in c# to declare jagged array that has two elements.

30) What is the difference between public, static and void?

- * You can access public declared variables anywhere in the application.
- * Static declared variables are globally accessible without creating an instance of the class.
- * Void is a type modifier that specifies that the method doesn't return any value.

31) What is ArrayList?

- * C# ArrayList is a non-generic collection.
- * The ArrayList class is defined in the System. Collections namespace.
- * Arraylist can store element of different data type
- * Arraylist can accept null values.
- * Arraylist allows the dynamic size and memory allocation.
- * Arraylist are not strongly typed.
- * You can add and remove the elements from an ArrayList at runtime.
- * also allows duplicate elements.

32) What is serialization?

- * serialization is the process that converting an object into stream of bytes.

- * meanse hum kisi object ko stream of bytes through kisi file, memory and database mai convert karke store kar sakte hai.
- * The reverse process of serialization is called deserialization.

33) difference between early binding and late binding in C#?

- * Early binding and late binding are the concept of polymorphism.
- * Compile Time Polymorphism: It is also known as early binding.
- * Run Time Polymorphism: It is also known as late binding.

34) Which are the access modifiers available in C#?

- * **1. Public:** If you define an attribute or method as public, it can be accessed from any code of the project.
- * **2. Private:** A private defined attribute or method can be accessed by any code within the containing class only.
- * **3. Protected:** If you define the method or attribute as protected it can be accessed by any method in the inherited classes and any method within the same class.
- * **4. Internal:** If you define an attribute or a method as internal, it is restricted to classes within the current position assembly.

35) What is delegate in C#?

- * A delegate in C# is an object that holds the reference to a method. It is like function pointer.
- * Delegates are mainly used in implementing the call-back methods and events.
- * Delegates can also be used in "anonymous methods" invocation.

36) What is Garbage Collection?

- * Garbage Collection is a process of releasing memory automatically occupied by objects which are no longer accessible.

37) Define managed and unmanaged code.

- * Managed code is developed using the Common Language Runtime (CLR) of the .NET framework. It can be directly executed in C#.
- * Unmanaged code is an unsafe code that doesn't run on CLR. It works outside of

the NET framework.

38) What is the difference between an Array and ArrayList in C#?

- * An array refers to the collection of similar variables under one name.
- * ArrayList is a collection of objects capable of being indexed separately.
- * The memory allocation is fixed in an array.
- * However, it can be increased or decreased in ArrayList.
- * The array has items with the same data type.
- * ArrayList can have items of different data types.

39) What Is AJAX in ASP.net?

- * AJAX in ASP.net stands for Asynchronous JavaScript and XML. This technique is used to develop rich web applications that are interactive, responsive, and dynamic in nature.
- * Ajax in ASP.net improves the responsiveness and interactivity of a website.
- * Ajax in ASP.net helps in reducing the traffic between server and client.
- * Ajax in ASP.net reduces the cross-browser issues as it is cross-browser friendly.

40) What is threading in C#?

- * Threading in C# is a way to run multiple tasks or pieces of code at the same time to make programs faster and more responsive.
- * A thread is a lightweight process that runs code independently. A program can have multiple threads running at the same time, each performing different tasks.
- * Every C# application starts with one thread, called the main thread. This is where the program's primary code runs.
- * Multithreading: This is the process of creating and running multiple threads simultaneously. It allows tasks to be performed in the background.

41) what is MVC in C#?

Model:

The Model represents the data and the business logic of the application. It handles retrieving, storing, and processing data.

View:

The View is responsible for displaying the user interface (UI) and presenting the data from the Model to the user.

Controller:

The Controller handles user input and requests. It processes the data from the Model and decides what View to render based on the user's actions.

42) What is ADO.NET?

- * ADO stands for Microsoft ActiveX Data Objects.
- * ADO.NET is one of Microsoft data access technologies, which we can use to communicate with different data sources.
- * It is a part of the .NET Framework, which connects the .NET Application and different data sources.
- * The Data Sources can be SQL Server, Oracle, MySQL, XML, etc.
- * Using the ADO.Net we can perform CRUD Operation.
- * Also we can create Console or Desktop Application

43) What is ASP.NET ?

- * the full form of ASP.Net is Active Server Page(NET - Network Enabled Technology).
- * ASP.Net was first released in the year 2002.
- * ASP.Net is a web API framework.
- * It is provided by Microsoft.
- * ASP.Net is a server side technology for web development.
- * Using the ASP.Net we can build dynamic websites, web applications, and web services.
- * ASP.NET provides tools, libraries, and a structure to make it easier to create websites and web apps.
- * ASP.NET has built-in features to handle security and performance optimization, which helps websites run faster and be more secure.

44) Hashtable Characteristics in C# (IMP).

- * The Hashtable Collection Class in C# stores the elements in the form of

key-value pairs.

- * Hashtable Class belongs to System.Collection namespace.
- * that is a Non-Generic Collection class.
- * It implements the IDictionary interface.
- * The Keys can be of any data type but they must be unique and not null.
- * The Hashtable accepts both null and duplicate values.
- * We can access the values by using the associated key.
- * The capacity of a Hashtable is the number of elements that a Hashtable can hold.
- * A hashtable is a non-generic collection, so we can store elements of the same type as well as of different types.

45) what is LINQ?

- * the full form of LINQ is language integrated query.
- * LINQ is uniform query syntax in C# and VB.NET to retrieve data from different sources and format.
- * In C#, LINQ is available in System.Linq namespace.
- * LINQ provides a query syntax similar to SQL, allowing developers to write queries using keywords like from, where, orderby, and select.
- * LINQ is strongly typed, ensuring compile-time checking and reducing runtime errors.
- * LINQ simplifies complex data queries, making code more readable and maintainable.
- * LINQ reduces the amount of code required for data manipulation and analysis.

46) What is JAVASCRIPT programming language?

- * JS is an interpreted, client-side, event-based, object-oriented scripting language.
- * Invented in 1995 at Netscape Corporation (LiveScript).
- * JS programs are run by an interpreter built into the user's web browser.
- * JS can dynamically modify an HTML page.
- * JS supports all browser.
- * JS can create cookies.
- * It is a case-sensitive language.
- * JS is not java programming language.

47) what is the Asp.Net page Life Cycle?

- * 1. Page is requested from server (sending request).
- * 2. Request and response object are created.
- * 3. Initialization of all page controls (Ex. Text, CheckBox, etc.).
- * 4. Page is loaded with default values.
- * 5. All validation controls here (checks form values).
- * 6. Event is triggered when same page is loaded again.
- * 7. All information sent to the user.
- * 8. Freeing up of memory.

48) What is Entity Framework in .Net Core ?

- * Entity Framework core is an ORM for .Net.
- * it allows developer to interact with databases using c# objects instead of SQL queries.
- * using the Entity Framework we achieve the database connection.
- * ORM simplifies the process of accessing data from Application.
- * Entity framework core is data access technology.
- * using the Entity framework we are able to talk with database.

49) Explain state management techniques in Asp.Net.

- * State management maintains and stores the information of any user till the end of user session.
- * ViewState: Stores data in the page itself.
- * Session State: Store data on the server per user session.
- * Application State: Store data on the server shared among all users.
- * Cookies: Store data on the client side.
- * Query String: Passes data in the URL.
- * Cache: Temporarily stores frequently accessed data in the server.

50) What are validators in ASP.NET?

- * RequiredField Validator: Check the input controller requirement (the field is not empty).
- * Range Validator: User values must be between two values.
- * Compare Validator: Compare the value of one input to another input.
- * Regular Expression Validator: Work on the specific pattern matching.
- * Custom Validator: Write your own validation controls.
- * Validation Summary: Give the validation error report.

51) What is Bundling?

* Bundling in ASP.NET is a way to combine multiple files (like CSS and JavaScript) into a single file to reduce the number of HTTP requests, thus improving load times.

52) What is Store Procedure?

* A stored procedure is a precompiled collection of SQL statements stored in a database.

* It can be executed with a single call, improving performance and security.

53) What is global.asax?

* global.asax is an optional file in ASP.NET that handles application-level events like Application_Start, Application_End, Session_Start, and Session_End.

54) ViewBag, ViewData, and TempData.

* **ViewBag**: Dynamic object for passing data from controller to view.

Easy to use because it's dynamic.

* **ViewData**: Dictionary object for passing data from controller to view. Works like a key-value pair.

Typecasting is required in ViewData.

* **TempData**: A special dictionary used to pass data between two requests.

55) what is Cache.

* Caching stores frequently accessed data in memory to improve performance and reduce database load. It can be used at various levels application.

56) What is a Generic Class?

* A generic class allows you to define a class with a placeholder for the data type, enabling type safety without sacrificing performance.

57) What is an Extension Method?

* An extension method allows you to add new methods to existing types without modifying them.

* It uses static methods and the this keyword.

58) What is routing in ASP.Net MVC?

- * Routing is used to handle incoming HTTP requests based on the URL.
- * ASP.Net MVC routing is a pattern matching system that is responsible for mapping incoming browser requests to specified MVC controller actions.

59) What is Master page in Asp.Net MVC ?

- * A Master Page is like a template that provides a common design or layout for all pages in your application.
- * It contains shared sections, like a header, footer, or navigation bar, which remain consistent across multiple pages.
- * Individual pages (child pages) inherit the design from the master page and only define the content specific to that page.
- * A Master Page Provides a consistent design/layout for multiple pages.

60) What is Dependency Injection in ASP.Net Core ?

- * Dependency injection is Software design pattern
- * that allows us to develop loosely coupled application.
- * Dependency injection is a process in which we are injecting a class of object to another class that depends on that object.

* Types of Dependency Injection

- 1) Constructor Injection:** Inject Dependencies through the class Constructor.
- 2) Property Injection:** Inject Dependency via a public Property.
- 3) Method Injection:** Passes dependency directly into a method.

61) What is Middleware in ASP.Net Core ?

- * In ASP.Net Middleware are two basic building blocks of the request processing pipeline.
- * Middleware in ASP.NET Core is a component that handles HTTP requests and responses in a pipeline.
- * Each middleware can perform actions like logging, security checks, or request modification, and can either pass control to the next middleware or stop the pipeline.
- * There are 3 functions used in Middleware that are 1) RUN 2) USE 3) MAP

62) What is REST Full API ?

- * REST Full API Stand for Representational State Transfer
- * REST Full API is a type of API that uses HTTP requests to access and use data.
- * It's lightweight and often used in web services.
- * REST API use Standard protocol used for communication on the internet.

63) What is Difference between ASP.Net & ASP.Net Core ?

- * .Net is support only windows operating system.
- * .Net Core support Cross Platform.
- * .Net Slower compared to Core.
- * .Net Core Faster than ASP.Net.
- * .Net is not open source.
- * ASP.Net Core us Fully open source.
- * .Net is Old Version no more update for .Net Framework.
- * .Net Core is New Updated Version.

64) What is JQuery in .Net ?

- * JQuery is library of Javascript.
- * JQuery Greatly Simplifies Javascript programming.
- * JQuery also provide easy DOM manipulation.
- * JQuery reduce the code Complaxsity.
- * JQuery increase the coding speed of developer.

65) What is DbContext Class ?

- * this class manage database connection.
- * it is used to retrive and save data from database.
- * this class related models and application.

66) IActionResult ?

- * IActionResult allows to return all types of result.
- * IHttpActionResult convert response to an HTTP message.

67) What is Outh ?

- * OAuth Stand For Open Authorization

- * OAuth is an open standard for authorization that allows third-party services to access a user's resources without sharing the user's credentials
- * Instead, it uses access tokens to grant secure, limited access.

68) Authentication and Authorization in .Net Core.

- * Process of verifying user's identity (checking username/password).
- * Example: Logging into Gmail using your email and password.
- * Process of giving permissions after authentication.
- * Example: In Gmail, only you can read your emails, not others.

Types Of Authentication and Authorization

- * Password-based Authentication → Login with username & password.
- * Multi-Factor Authentication (MFA) → Combination (password + OTP/email/mobile).
- * Biometric Authentication → Fingerprint, face, retina scan.
- * Token-based Authentication → Uses tokens (like JWT) after login.
- * Certificate-based Authentication → Uses digital certificates.
- * Single Sign-On (SSO) → Login once, access multiple apps.
- * OAuth / OpenID Connect → Login via third-party (Google, Facebook, etc.).

Section of SQL

69) what are joins in SQL?

- * **INNER JOIN:** Return matching records between two tables.

Syntax:

```
SELECT column_name(s)
FROM TableA
INNER JOIN TableB
ON TableA.col_name = TableB.Col_name
```

- * **LEFT JOIN:** Return all data of Left table and also return matched data from Right table.

Syntax:

```
SELECT column_name(s)
FROM TableA
```

LEFT JOIN TableB
ON TableA.col_name = TableB.Col_name

* **RIGHT JOIN:** Return all data of Right table and also return matched data from Left table.

Syntax:

SELECT column_name(s)
FROM TableA
RIGHT JOIN TableB
ON TableA.col_name = TableB.Col_name

* **FULL JOIN:** Return all data from both table Right Table or Left Table.

Syntax:

SELECT column_name(s)
FROM TableA
FULL JOIN TableB
ON TableA.col_name = TableB.Col_name

70) What is Primary key in SQL?

* A Primary key is a column or a group of columns in a table that uniquely identifies the rows of data in that table.

* Primary key is the combination of a NOT NULL and UNIQUE.

EX. Unique + Not Null

71) What is a foreign key?

* The foreign key is used to link one or more tables together.

* foreign key field in one table refers to the primary key field of the other table.

* It identifies each row of another table uniquely that maintains the referential integrity.

72) What is a unique key?

* Unique Key is Unique Identifier

* We can store NULL value in the unique key column, but only one NULL is allowed.

* We can modify the unique key column values.

73) SQL Commands.

* **DDL** - Data Definition Language.

EX. Create, Alter, Drop.

* **DML** - Data Manipulation Language.

EX. Select, Insert, Update, Delete.

* **DCL** - Data Control Language.

EX. Grant, Revoke.

* **TCL** - Transaction Control Language

EX. Commit, Rollback, Savepoint.

74) What is a "TRIGGER" in SQL?

* TRIGGER is work like Recycle Bin.

* SQL triggers have two main components one is action, and another is an event.

* when user delete a record from a table then the record store in bacup table.

75) What is the ACID property in a database?

* The ACID properties are meant for the transaction that goes through a different group of tasks.

* Atomicity

* Consistency

* Isolation

* Durability

76) What us difference between Delete, Deop and Truncate?

* **Delete** : Removes specific rows from a table (using WHERE).

* **Truncate** : Removes all rows from a table (cannot use WHERE).

* **Drop** : Deletes the entire table (structure + data).

77) What is difference between DBMS and RDBMS in SQL.

DBMS	RDBMS
<u>DBMS</u> stores data as file.	<u>RDBMS</u> stores data in tabular form.
Data elements need to access individually.	Multiple data elements can be accessed at the same time.
No relationship between data.	Data is stored in the form of tables which are related to each other.
Normalization is not present.	Normalization is present.
DBMS does not support distributed database.	RDBMS supports distributed database.
It stores data in either a navigational or hierarchical form.	It uses a tabular structure where the headers are the column names, and the rows contain corresponding values.
It deals with small quantity of data.	It deals with large amount of data.
Data redundancy is common in this model.	Keys and indexes do not allow Data redundancy.
It is used for small organization and deal with small data.	It is used to handle large amount of data.
Not all Codd rules are satisfied.	All 12 Codd rules are satisfied.
Security is less	More security measures provided.
It supports single user.	It supports multiple users.
Data fetching is slower for the large amount of data.	Data fetching is fast because of relational approach.

78) What is the difference between the WHERE and HAVING clauses?

WHERE	HAVING
This clause is implemented in row operations.	This clause is implemented in column operations.
It does not allow to work with aggregate functions.	It can work with aggregate functions.
This clause can be used with the SELECT, UPDATE, and DELETE statements.	This clause can only be used with the SELECT statement.

API SECTION

79) What is an API?

- * API stands for Application Programming Interface.
- * It's a set of rules and tools that allow different software applications to communicate with each other.
- * The main use of an API is to allow one program to interact with another.
- * Websites use APIs to fetch data like weather reports, stock prices, or even social media posts.
- * For example, if you're using a weather app, the app uses an API to get weather data from a server.

80) Types of APIs.

- * Open APIs (Public APIs): These are available to any developers, usually for free or with a subscription, like Google Maps API.
- * Internal APIs (Private APIs): These are used within a company to connect internal systems.
- * Partner APIs: These are shared with specific business partners to provide services.
- * Composite APIs: These combine multiple APIs to work together and return one set of data.

81) What is RESTful API?

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- * It's lightweight and often used in web services.
- * REST Full API Use Standard protocol for communication on the internet.

82) What is the difference between REST and SOAP?



RESTful API

- **Full form:** Representational State Transfer
- **Format:** Works with JSON, XML, or plain text (mostly JSON).
- **Style:** Lightweight, faster, easy to use.
- **Transport:** Mostly uses HTTP (GET, POST, PUT, DELETE).
- **Flexibility:** Can be used on web, mobile, IoT easily.
- **Best for:** Modern web apps, mobile apps, speed & scalability.



SOAP API

- **Full form:** Simple Object Access Protocol
- **Format:** Always uses XML (strict structure).
- **Style:** Heavier, more complex.
- **Transport:** Can use HTTP, SMTP, TCP (not just HTTP).
- **Security:** Has built-in standards (WS-Security, ACID).
- **Best for:** Banking, payment gateways, enterprise apps

83) What is HTTP and how is it related to APIs?

* HTTP (HyperText Transfer Protocol) is the foundation of data communication for the web. Most APIs, especially RESTful ones, use HTTP to send and receive data.

84) What is the difference between API and Web Service?

- * API: A set of rules for two software programs to communicate.
- * Web Service: A specific kind of API that operates over a network (usually the internet).

85) What is API versioning?

* API versioning allows developers to make changes to an API without breaking old versions. Different versions ensure compatibility for existing users.

86) What is API testing?

* API testing checks if an API is working correctly by sending requests and verifying the responses. It ensures the API meets the expected behavior.

87) What are API Services in .Net Core

1. Routing Service

- * Decides which controller and action will handle a request.
- * Example: /api/products ? goes to ProductsController.

2. Controller Service

- * Main part that receives the request and processes it.
- * Acts like the "brain" of API.

3. Validation Service

- * Checks if data is valid before processing.
- * Example: Age can't be negative, email must be in correct format.

4. Dependency Injection (DI) Service

- * Provides required classes/services to controllers.
- * Makes code more flexible and testable.

88) what is IEnumerable in .Net

- * `Enumerable` is an interface in C#.
- * It does not store data itself, but collections like `Array`, `List`, etc. implement it.
- * It works with the `foreach` loop.
- * It allows us to loop through elements one by one in a collection.

89) What is CLR in C# ?

- * CLR stands for **Common Language Runtime** in .NET.
- * It is the **heart of .NET Framework** (runtime environment).
- * CLR **executes the code** written in C#, VB.NET, F#, etc.
- * It provides services like **memory management, garbage collection, exception handling, and security**.
- * CLR converts **IL code (Intermediate Language)** into **machine code** using JIT (Just-In-Time compiler).

90) what is async/await in ASP.Net Core ?

- * `async` and `await` are used for **asynchronous programming** in C#.
- * They allow code to run **without blocking the main thread**.
- * `async` keyword marks a method as asynchronous.
- * `await` keyword pauses the method until the awaited task is finished.

91) What are status codes in API?

* Status codes indicate the result of a request to an API. For example:

✓ Success Codes

- **200 OK** → Request worked fine.
- **201 Created** → New resource created successfully.
- **202 Accepted** → Request accepted, still processing.
- **204 No Content** → Request worked but no data to return.

✗ Client Error Codes (problem from user side)

- **400 Bad Request** → Something wrong in request (syntax/invalid input).
- **401 Unauthorized** → User not logged in / missing authentication.
- **403 Forbidden** → User logged in but not allowed.
- **404 Not Found** → Resource/URL doesn't exist.
- **405 Method Not Allowed** → Wrong HTTP method (e.g., used GET instead of POST).
- **409 Conflict** → Request conflicts with current state (e.g., duplicate data).
- **429 Too Many Requests** → User sent too many requests (rate limit).

🚨 Server Error Codes (problem from server side)

- **500 Internal Server Error** → General server error.
- **501 Not Implemented** → Feature not available on server.
- **502 Bad Gateway** → Server got invalid response from another server.
- **503 Service Unavailable** → Server is down or busy.
- **504 Gateway Timeout** → Server didn't respond in time.

92) Explain Value Types vs Reference Types in C#.

☑ Value Types in C#

- * Stored **directly in memory (stack)**.
- * Hold the actual **data**.
- * Each variable has its **own copy**.
- * Examples: `int`, `float`, `bool`, `char`, `struct`, `enum`.

☑ Reference Types in C#

- * Stored in **heap memory**, variable holds a **reference (address)**.
- * Hold the **memory address** where data is stored.
- * Multiple variables can **point to the same object**.
- * Examples: `string`, `class`, `array`, `object`, `interface`.

93) what is Exeption handling in .net Core ?

- * **Exception Handling** means catching and handling those errors so the program doesn't crash.
- * We use **try, catch, finally** blocks to handle exceptions.
- * **try** → code that might throw an error.
- * **catch** → code to handle the error.
- * **finally** → code that always runs (like cleanup).
- * We can also use **throw** to re-throw an exception.

94) Explain JWT Authentication & Authorization in AP_{Is}

- * JWT Stand For "JSON Web Token" Authentication & Authorization
- * **JWT** is a secure way to pass user identity between client and server.
- * It is a **token (string)** created after a user logs in successfully.
- * The token has 3 parts: **Header, Payload (user data), Signature**.
- * JWT is **stateless** → server does not need to store user sessions

95) What are Action Filters in .Net Core ?

- * **Action Filters** are special attributes in ASP.NET Core MVC.
- * They run **before or after** a controller action executes.
- * Used for **cross-cutting tasks** like logging, caching, validation, error handling, authorization.
- * We apply them using attributes like `[Authorize]`, `[ValidateAntiForgeryToken]`, `[HttpGet]`, etc.
- * You can also create **custom action filters** by inheriting from `ActionFilterAttribute`.

96) what is Microservices in .NET Core ?

- * **Microservices** = breaking a big application into **small, independent services**.
- * Each microservice focuses on a **single business function** (e.g., Payment, Order, User).
- * Microservices **communicate with each other** using APIs (mostly REST or gRPC).
- * In .NET Core, we build microservices using **ASP.NET Core Web API**.
- * Each microservice can have its **own database, technology, and deployment**.
- * Benefits → **Scalable, flexible, easy to deploy, independent updates**.

97) What is Difference between Lazy Loading vs Eager Loading in EF Core.

☒ Lazy Loading in EF Core

- * Data is loaded **only when it is accessed** (on demand).
- * First, only main entity is fetched → related data fetched later when needed.
- * Saves memory if related data is **not always required**
- * Can cause **multiple database calls** (performance issue if used a lot).

☒ Eager Loading in EF Core

- * Data is loaded **at the same time** as the main entity.
- * Uses **Include ()** method to fetch related data.
- * Good when you **know you'll need related data**.
- * Only **one database call**, but can load extra unused data.

98) What are Migrations in EF Core?

- * **Migrations** are a way to keep your **database schema** (tables, columns) in sync with your **C# models**.
- * When you change a model (add/remove property), migrations update the database accordingly.
- * They create **C# migration files** that contain instructions to apply changes.

Common commands :

- * **Add-Migration** <name> → creates migration file.
- * **Update-Database** → applies migration changes to the database.

99) What are Indexes and how do they improve performance in SQL?

- * An **Index** is like a **table of contents** in a book → helps you find data faster.
- * It is created on one or more columns of a table.
- * Without index → SQL scans the **entire table** (slow if table is big).
- * With index → SQL directly goes to the matching row (much faster).

100) What is Data Binding?

- * **Data Binding** means connecting a UI element (like textbox, label, grid) to a **data source** (like variable, object, database).
- * It helps display and update data **automatically** between UI and data source.

☒ **Types of Data Binding**

- * **One-Way Binding** => Data flows **from source** → UI.
- * **Two-Way Binding** => Data flows **both ways** (source ↔ UI).
- * **One-Time Binding** => Data is set **only once** when UI is loaded.
- * **One-Way to Source Binding** => Data flows **from UI** → source only.